Biology Class 11 National Book Foundation FBISE Quality Notes
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Chapter 13: Chromosome and DNA

13.1 Chromosome structure (SLO 1)

- ***Chromosomes*** *(Gk chroma = colour, soma = body; named by* ***Waldeyer****, 1888) are thread-like structures in the nucleus that take basic stains. Each is made of* ***DNA and proteins*** *(about 40% DNA + 60% protein); one chromatid = one DNA molecule coiled with histones. Visible only in cell division; elsewhere uncondensed as* ***chromatin****.*

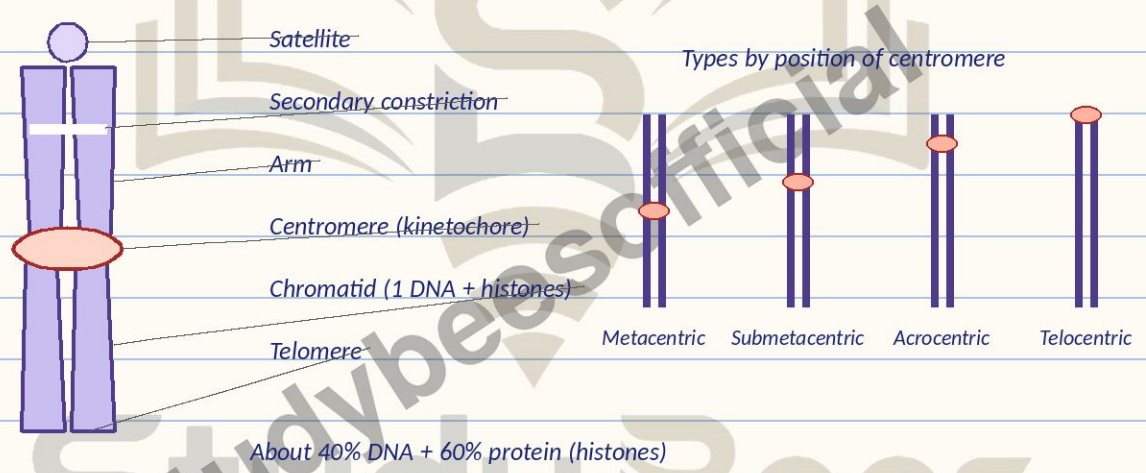


Fig: Structure and types of chromosome

- ***Centromere*** *(primary constriction; kinetochore for spindle attachment) divides a chromosome into two* ***arms****.* ***Telomeres*** *(chromosome ends) protect against degradation and fusion.* ***Secondary constriction*** *(nucleolar organizer, makes rRNA) may form a* ***satellite****. Types by centromere position:* ***metacentric*** *(middle, V-shaped at anaphase),* ***submetacentric*** *(L-shaped),* ***acrocentric*** *(near end, J-shaped),* ***telocentric*** *(at the tip, rod). Prokaryotic chromosome is a single circular DNA without histones (plasmids extra).*
- ***Euchromatin:*** *lightly packed, genes active;* ***heterochromatin:*** *tightly packed,*

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genes inactive.

- **Nucleosome:** DNA wrapped twice around a core of **8 histone proteins** (two each of H2A, H2B, H3, H4); H1 binds linker DNA; about 200 nucleotide pairs per nucleosome: "beads on a string".

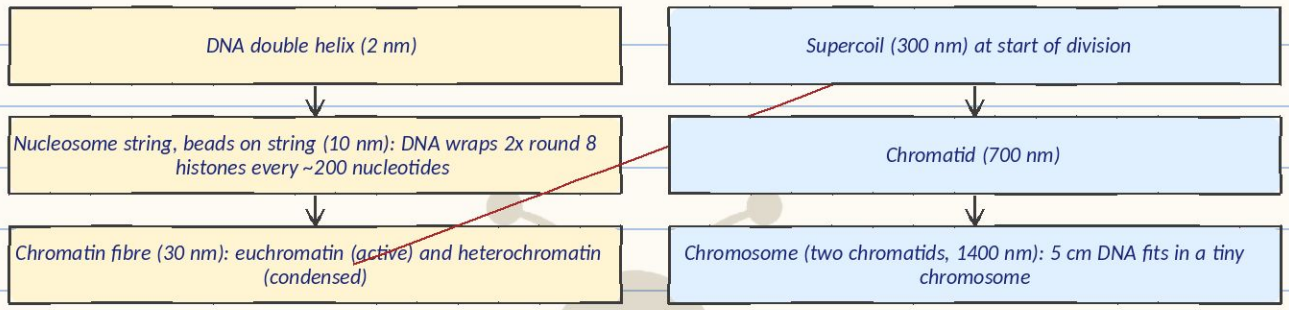
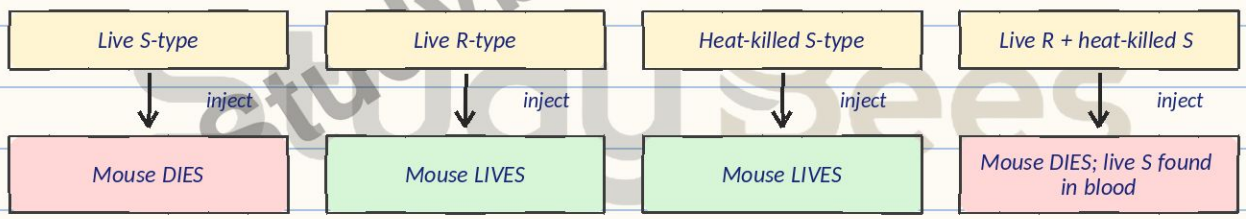


Fig: Levels of organization of a chromosome

13.2 DNA as the genetic material (SLO 2)

- Proteins (20 amino acids) were once thought to be the genetic material; **Miescher** (1869) found nucleic acid (nuclein) in pus cells. Evidence for DNA:



S-type: smooth, polysaccharide capsule, virulent; R-type: rough, no capsule, non-virulent

Transformation: heat-killed S passed genetic material to live R (turned it into S)

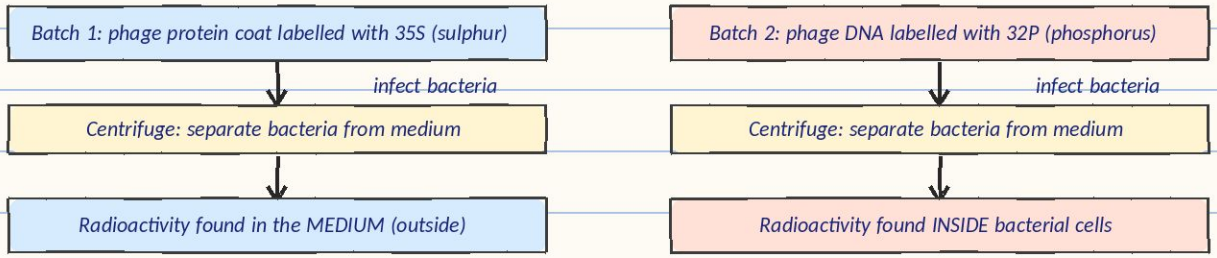
Avery, Macleod, McCarty (1944): only DNAase stops transformation = DNA is the transforming principle

Fig: Griffith's experiment (1928) on Streptococcus pneumoniae

- **Griffith** (1928, **Streptococcus pneumoniae**): smooth (S, capsule, virulent) kills mice; rough (R, non-virulent) does not; heat-killed S does not; **live R + heat-killed S** kills mice and live S recovered: **transformation** by a "transforming principle". **Avery, MacLeod and McCarty** (1944): the principle is DNA (DNAase

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stopped transformation).



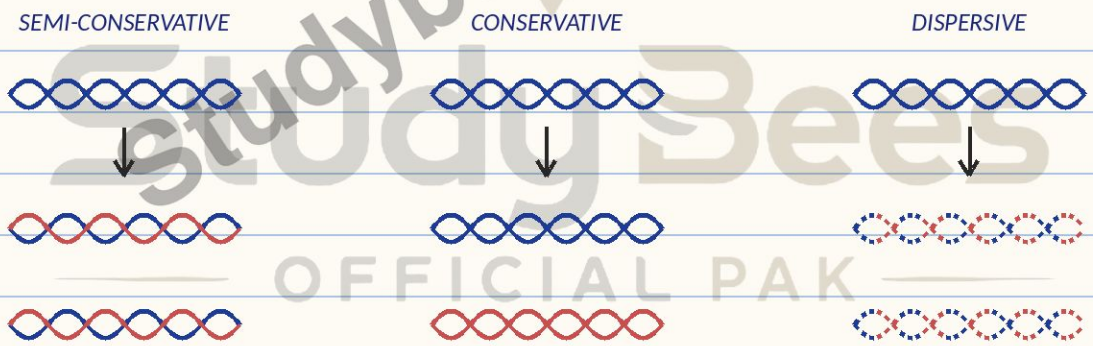
Conclusion: DNA (not protein) enters the cell and directs production of new viruses (1952)

Fig: Hershey and Chase experiment on bacteriophage

- ***Hershey and Chase** (1952): T2 bacteriophage proteins labelled with ^{35}S, DNA with ^{32}P; only ^{32}P entered the bacteria: DNA is the genetic material. Other evidence: DNA amount constant in somatic cells and half in gametes; mutations in DNA change characters; UV is absorbed by DNA and mutates.*

13.3 DNA replication (SLO 3-6)

- *DNA makes an identical copy during the **S phase**, before cell division. Three models were proposed:*



Blue = parental strand; red = newly made strand

Meselson and Stahl (1958) proved the semi-conservative model

Fig: Three models of DNA replication

Model	Description
Semi-conservative	Each daughter DNA has one old (parental) strand and one new strand (Watson and Crick suggestion).
Conservative	Parental DNA stays intact; the daughter DNA is wholly new.

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Model	Description
Dispersive	Parental and new DNA mixed in fragments in each strand.

Meselson-Stahl experiment (1958)

- *E. coli grown for many generations in **heavy nitrogen (^{15}N)** medium (heavy DNA), then shifted to light ^{14}N medium. DNA from generations extracted and run in a **CsCl density gradient** (ultracentrifuge). After one generation: one **intermediate (hybrid)** band: excludes the conservative model. After two generations: one hybrid band + one light band: excludes the dispersive model. Result: **semi-conservative replication**.*

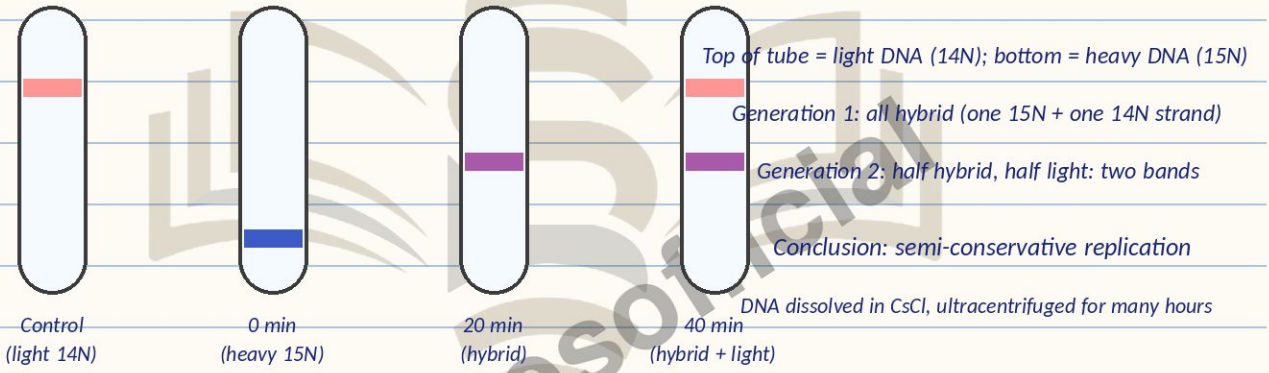
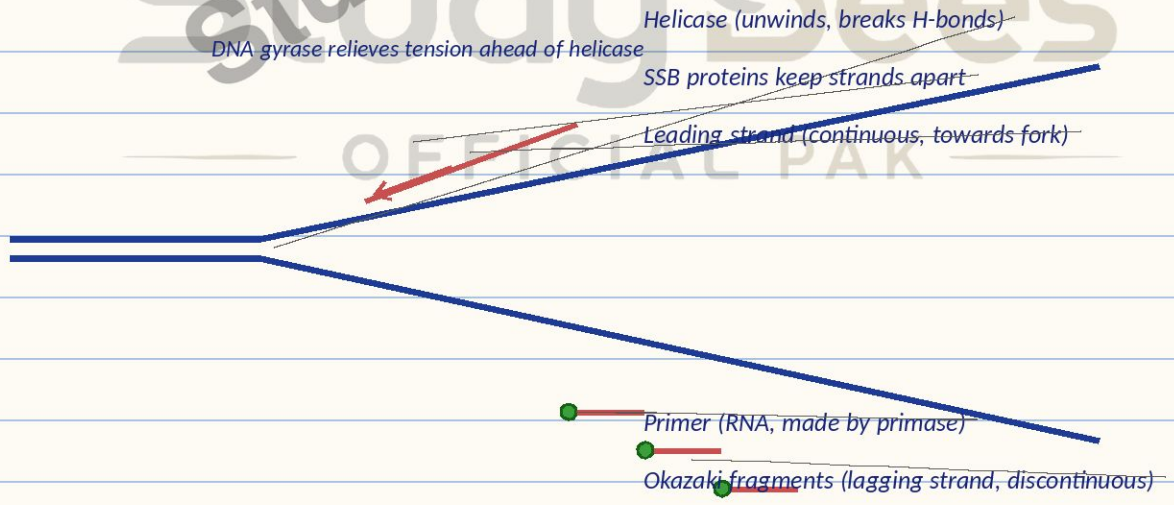


Fig: Meselson-Stahl experiment (heavy nitrogen 15N shifted to 14N medium)

Events of replication (SLO 5)



DNA polymerase III: adds nucleotides 5' to 3' on 3'-OH of primer; proofreading

DNA polymerase I replaces primers by DNA; DNA ligase joins fragments (phosphodiester bonds)

Fig: Replication fork

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1. ***Origin:*** *replication starts at the* ***origin of replication****; prokaryotes have one origin, eukaryotes many (replication bubbles); synthesis proceeds in* ***both directions*** *(bidirectional) forming two* ***replication forks****.*
2. ***Unwinding: helicase*** *separates strands by breaking hydrogen bonds;* ***single-strand binding proteins*** *keep them apart;* ***DNA gyrase / topoisomerase*** *relieves supercoil tension.*
3. ***Priming: primase*** *makes a short* ***RNA primer*** *on each template; DNA polymerase needs a free 3'-OH.*
4. ***Elongation: DNA polymerase III*** *adds nucleotides only in the* ***5' to 3'*** *direction (reads template 3' to 5').* ***Leading strand:*** *continuous towards the fork.* ***Lagging strand:*** *discontinuous short pieces,* ***Okazaki fragments****, each with its own primer.*
5. ***Primer removal: DNA polymerase I*** *removes RNA primers and fills with DNA;* ***DNA ligase*** *joins the fragments by phosphodiester bonds.*
6. ***Proofreading:*** *polymerases check and correct mismatches (3' to 5' exonuclease); error is about one in* 10^9 *nucleotides.*

- ***Stability and variability of DNA (SLO 6):*** *replication is accurate by complementary pairing (A-T, G-C), proofreading and repair, so DNA is stable; occasional errors and mutations provide* ***variability****. A balance of both is needed.*

13.4 Genetic code (SLO 7, 8)

Genetic code: *The set of rules by which the nucleotide sequence of DNA / mRNA is translated into the amino acid sequence of a protein. Code = the relationship; codon = a triplet of mRNA bases.*

- ***Codon*** *= three consecutive mRNA nucleotides = one amino acid. Four bases give* 4^3 *=* ***64 codons****:* ***61 code for the 20 amino acids*** *and* ***3 are stop codons*** *(UAA, UAG, UGA);* ***AUG*** *= start codon (methionine).* ***Crick and Brenner*** *showed a triplet code. Features:* ***triplet; non-overlapping; comma-less (continuous); degenerate*** *(several codons for one amino acid, e.g. 6 for leucine; but one*

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codon codes only one amino acid: unambiguous); ***universal*** *(nearly all organisms).*

- ***Nirenberg and Matthaei*** *(1961) broke the first codon: poly-U gave polyphenylalanine (UUU = Phe).* ***Khorana*** *made synthetic mRNAs;* ***Holley*** *described tRNA; Nobel Prize 1968.* ***Wobble:*** *the third base of a codon pairs loosely with the anticodon.*

13.5 Transcription (SLO 9, 10)

Transcription: *The synthesis of RNA (mRNA, rRNA, tRNA) on a DNA template by the enzyme* ***RNA polymerase****; no primer is needed. Only the* ***template (antisense)*** *strand is read; the other is the coding (sense) strand.*

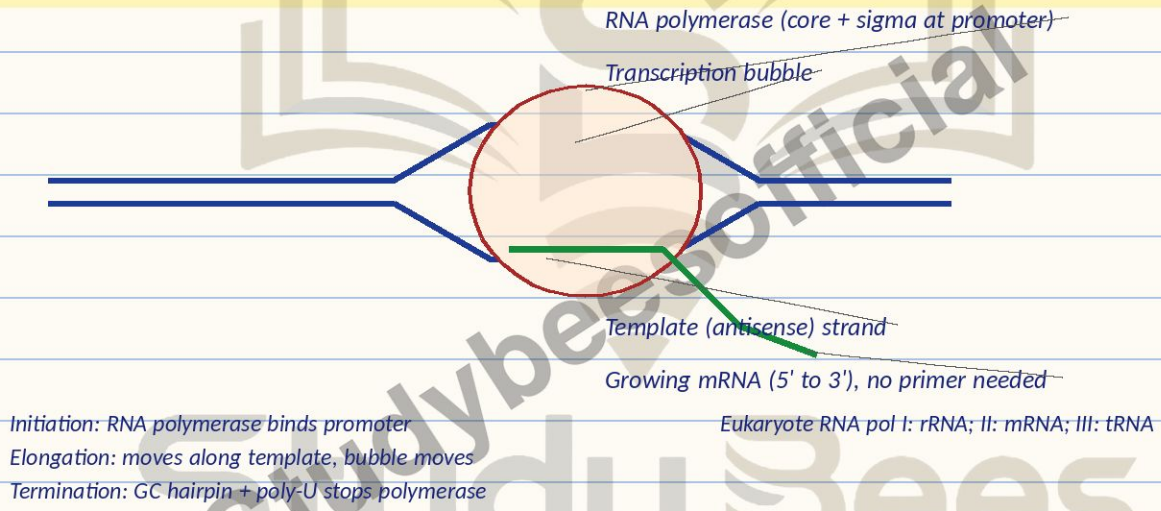


Fig: Transcription (synthesis of RNA from DNA)

1. ***Initiation:*** *RNA polymerase (prokaryote: core enzyme + sigma factor) binds the* ***promoter*** *(TATA box about -25, CAAT box about -80; prokaryote -10 and -35 boxes) and opens the DNA (bubble).*
2. ***Elongation:*** *RNA polymerase adds ribonucleotides 5' to 3' (A-U, T-A, G-C, C-G), the bubble moves along.*
3. ***Termination:*** *at the terminator (a GC-rich hairpin followed by poly-U) the enzyme stops; mRNA is released.*

- *Eukaryotes: RNA polymerase I (rRNA), II (mRNA), III (tRNA, small RNA).*

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Prokaryotic mRNA is ready at once and translated even while being made; ***polycistronic****.*

mRNA processing - why eukaryotic mRNA is shorter (SLO 10)

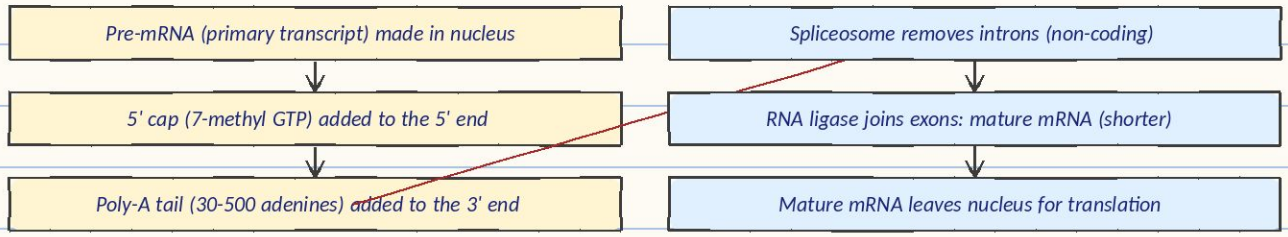


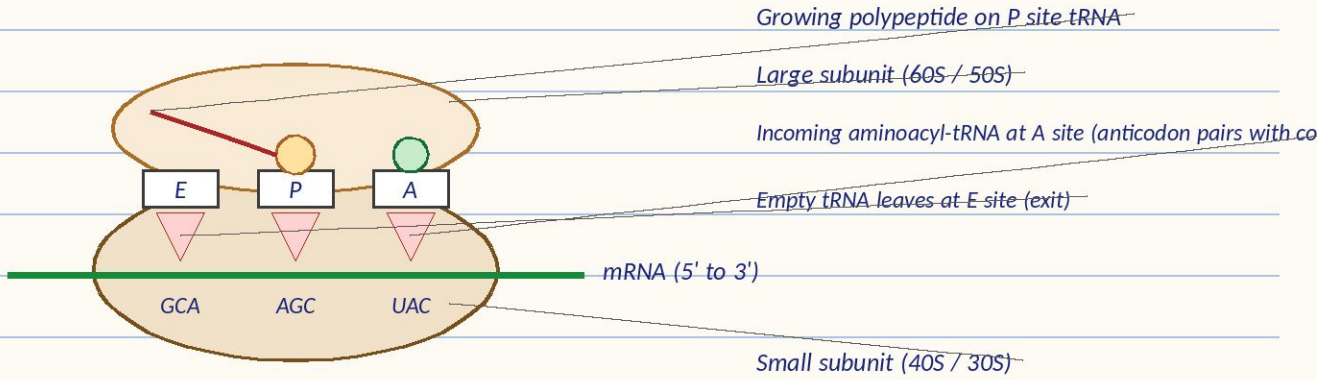
Fig: mRNA processing in eukaryotes (why mRNA shortens)

- *Eukaryotic genes are* ***split****:* ***exons*** *(coding) and* ***introns*** *(non-coding). The primary transcript (****pre-mRNA / hnRNA****) is processed:* ***5' cap****,* ***poly-A tail*** *(stability and export), and* ***splicing*** *by the* ***spliceosome*** *(snRNPs) which removes introns and joins exons. Hence the mature mRNA is much shorter than the gene.* ***Alternative splicing*** *gives different proteins from one gene.*

13.6 Translation - protein synthesis (SLO 11-13)

- ***Ribosomes*** *(rRNA + proteins; 80S eukaryote, 70S prokaryote) have two subunits and three tRNA sites:* ***A (aminoacyl), P (peptidyl), E (exit)****.* ***tRNA*** *(cloverleaf) carries an amino acid at the 3' CCA end and an* ***anticodon*** *pairing with the mRNA codon.* ***Aminoacyl-tRNA synthetase*** *attaches the correct amino acid using ATP (activation).*

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Elongation steps: (a) aminoacyl-tRNA enters A site; (b) peptidyl transferase makes peptide bond; (c) translocation moves ribosome 5' to 3' (A to P, P to E). Stop codon UAG, UAA or UGA: release factors.

Fig: Translation - polypeptide elongation on the ribosome

1. ***Initiation:*** *small subunit binds mRNA at the* ***start codon AUG****; initiator tRNA (Met) pairs; large subunit joins; initiation factors and GTP. (Prokaryotes: Shine-Dalgarno sequence; fMet; eukaryotes: 5' cap.)*
2. ***Elongation:*** *aminoacyl-tRNA enters the A site;* ***peptidyl transferase*** *forms a peptide bond; ribosome moves one codon (****translocation****), tRNA shifts A to P to E and leaves; cycle repeats.*
3. ***Termination:*** *a stop codon (UAA, UAG, UGA) enters the A site;* ***release factors*** *free the polypeptide; ribosome subunits separate. Many ribosomes read one mRNA =* ***polysome****.*

Prokaryotic translation	***Eukaryotic translation***
In cytoplasm, coupled with transcription (no nucleus)	*In cytoplasm after mRNA leaves the nucleus*
70S ribosome (30S + 50S)	*80S ribosome (40S + 60S)*
First amino acid formylmethionine; Shine-Dalgarno sequence	*First amino acid methionine; 5' cap recognized*
Polycistronic mRNA; no processing	*Monocistronic mRNA processed (cap, tail, introns removed)*

- ***Fate of the polypeptide (SLO 13):*** *the chain folds into secondary / tertiary structure with help of* ***chaperones****, may be cleaved or modified (glycosylation, phosphorylation). Proteins made on free ribosomes stay in the cytosol or enter*

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*nucleus, mitochondria; proteins made on ribosomes of **rough ER** enter the ER lumen, move via vesicles to the **Golgi** for packaging and are secreted or sent to lysosomes or the membrane (signal sequence).*

13.7 Regulation of gene expression (SLO 14-16)

- *Not all genes work all the time. **Gene regulation** makes the right protein in the right cell at the right time, saves energy and gives differentiation. Levels: transcriptional (most important), RNA processing, translational, post-translational.*
- ***Jacob and Monod** (1961, E. coli): **lac operon**. An **operon** = group of structural genes with common **promoter** and **operator**, controlled together. **Lac operon: i gene** (regulator) makes the **repressor**; promoter; operator; structural genes **z** (beta-galactosidase), **y** (permease), **a** (transacetylase).*

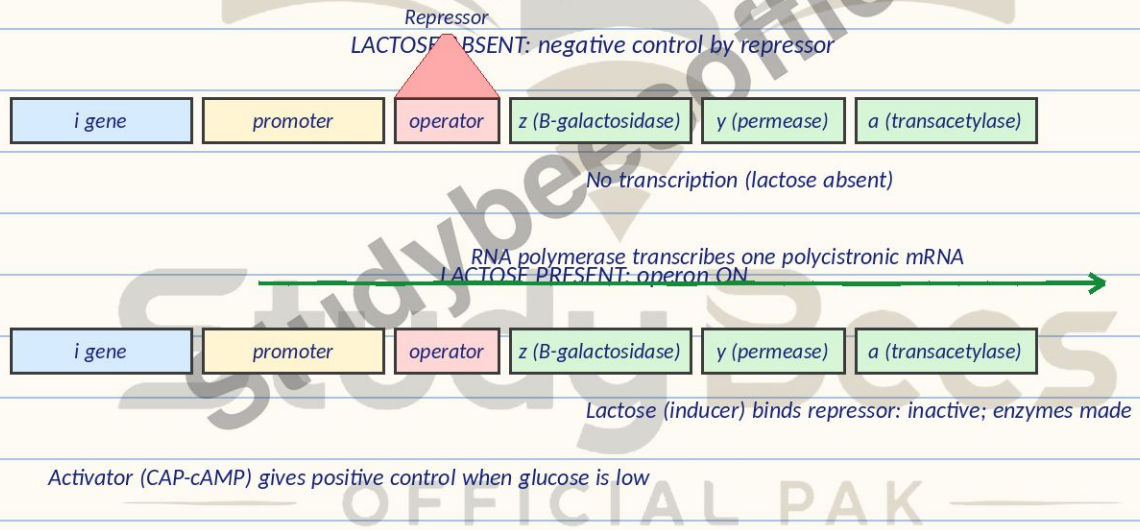


Fig: Lac operon (positive and negative control of gene expression)

- ***Negative control (SLO 15):** no lactose: repressor binds the **operator**, blocks RNA polymerase: structural genes off. With lactose (**inducer**, actually allolactose): it binds the repressor, which leaves the operator: genes transcribed, enzymes digest lactose; **inducible** operon. **Positive control (SLO 16):** when glucose is low, cAMP binds **CAP** (catabolite activator protein) which attaches near the promoter and **increases** transcription; glucose present: low*

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cAMP, operon stays low (catabolite repression).

- *Eukaryotes use enhancers, silencers, transcription factors, chromatin changes (methylation, acetylation), miRNA.*

13.8 Mutations (SLO 17-22)

***Mutation:** A sudden, permanent, heritable change in the DNA base sequence or chromosome number / structure of an organism. **Hugo de Vries** coined the term (evening primrose). The changed organism is a **mutant**; the causative agent a **mutagen**.*

- ***Sources: spontaneous (natural):** errors of replication, tautomeric shifts, transposons, spontaneous chemical change, background radiation. **Induced: physical mutagens** (UV, X-rays, gamma rays, high temperature), **chemical mutagens** (nitrous acid, mustard gas, base analogues, benzpyrene, formaldehyde, tobacco tar), **biological** (some viruses).*

Natural mutation	Induced mutation
Occurs spontaneously (error in replication, background radiation)	Caused by mutagens (radiation, chemicals)
Rate low (about 1 in 10^5 to 10^9)	Rate high

Gene (point) mutations (SLO 22)

Normal mRNA	AUG AGC GCA UUC	Met - Ser - Ala - Phe
Substitution	AUG AGC GCU UUC	Met - Ser - Ala - Phe (silent) or changed amino acid
Insertion	AUG AGC AGC AUU C	Frameshift: all codons after change
Deletion	AUG AGC CAU UC	Frameshift: non-functional protein

Fig: Types of point mutation (sample sequence)

Point mutation = base substitution, insertion or deletion; frameshift mutation = insertion / deletion not in multiples of three

- ***Substitution:** one base replaced: **silent** (same amino acid), **missense** (different amino acid; e.g. sickle cell: GAG to GUG, glutamic acid to valine, in beta-globin), **nonsense** (becomes a stop codon). **Frameshift:** insertion or deletion of bases (not multiple of three) shifts the reading frame, changing*

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*every later amino acid. **Phenylketonuria (PKU):** mutation in gene for phenylalanine hydroxylase; phenylalanine accumulates: brain damage unless diet is controlled. Also albinism, haemophilia, cystic fibrosis.*

Chromosomal mutations (SLO 21)

- ***Structural: deletion** (segment lost), **duplication** (repeated), **inversion** (segment reversed), **translocation** (segment moved to non-homologue). **Numerical: aneuploidy** by **non-disjunction** (failure of homologues / chromatids to separate in meiosis): monosomy or trisomy; **polyploidy** (extra complete sets, common in plants, e.g. wheat).*

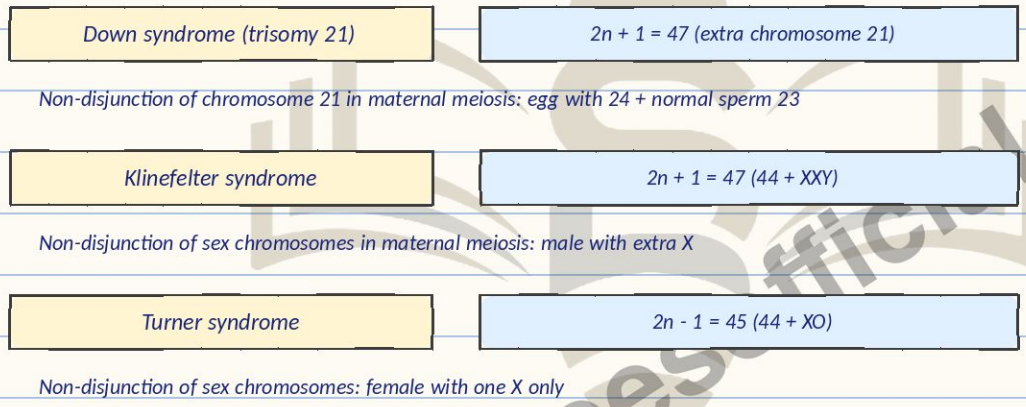


Fig: Chromosomal mutations through non-disjunction

- ***Down syndrome:** trisomy 21 (47 chromosomes); short, learning difficulty, slanting eyes, risk rises with mother's age. **Klinefelter:** XXY male, sterile, tall, breast development. **Turner:** XO female (45), short, sterile, no ovaries. **Cri-du-chat:** deletion of part of chromosome 5.*

Importance of mutations (SLO 19, 20)

- ***Most mutations are harmful** (recessive, deleterious): a random change in a finely tuned protein usually lowers function (e.g. enzyme destroyed, cancer). Few are **neutral** (silent) and a very few **beneficial**.*
- ***Mutations and evolution:** mutations are the ultimate source of **new alleles / variation**; if beneficial under a given environment they are favoured by natural selection and spread (e.g. antibiotic resistance, industrial melanism); so evolution and adaptation depend on them. Also used in plant breeding*

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(induced mutations; Pakistan's NIAB mutant cotton, rice, wheat varieties).

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Chapter 13: Short Questions with Answers (SLO based)

Q1. [SLO 1] What is a chromosome? Who named it?

Ans: A thread-like structure of DNA and histone proteins in the nucleus visible in cell division; named by Waldeyer (1888).

Q2. [SLO 1] Differentiate chromatin and chromosome.

Ans: Chromatin: thin uncoiled DNA-protein threads in non-dividing nucleus. Chromosome: condensed form in cell division.

Q3. [SLO 1] What are the parts of a chromosome?

Ans: Two chromatids, centromere (kinetochore), arms, telomeres and sometimes secondary constriction and satellite.

Q4. [SLO 1] What is a telomere? A centromere?

Ans: Telomere: end of chromosome protecting it. Centromere: primary constriction where spindle fibres attach.

Q5. [SLO 1] Classify chromosomes by position of centromere.

Ans: Metacentric (middle), submetacentric, acrocentric (near end) and telocentric (at tip).

Q6. [SLO 1] Differentiate euchromatin and heterochromatin.

Ans: Euchromatin: lightly packed, genes active. Heterochromatin: tightly packed, genes inactive.

Q7. [SLO 1] What is a nucleosome?

Ans: DNA wound twice around a core of 8 histone proteins (about 200 base pairs); beads on a string.

Q8. [SLO 1] What are histones?

Ans: Basic proteins (H2A, H2B, H3, H4, H1) around which DNA is wound.

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Q9. [SLO 1] Describe the stages of DNA packaging into a chromosome.

Ans: Double helix (2 nm), nucleosome string (10 nm), chromatin fibre (30 nm), supercoil (300 nm), chromatid (700 nm), chromosome (1400 nm).

Q10. [SLO 2] What is transformation? Who discovered it?

Ans: Change in the genotype of a bacterium by taking up DNA from another; discovered by Griffith (1928).

Q11. [SLO 2] Differentiate S and R strains of Streptococcus pneumoniae.

Ans: S: smooth, capsule, virulent (kills mice). R: rough, no capsule, non-virulent.

Q12. [SLO 2] What did Avery, MacLeod and McCarty find?

Ans: In 1944 they showed DNA is the transforming principle (DNAase stopped transformation).

Q13. [SLO 2] What was the aim and result of the Hershey-Chase experiment?

Ans: To find whether DNA or protein is genetic material; ^{32}P labelled DNA entered bacteria, ^{35}S protein stayed outside, proving DNA is genetic material.

Q14. [SLO 2] Why was ^{35}S used for protein and ^{32}P for DNA?

Ans: Protein contains sulphur but no phosphorus; DNA has phosphorus but no sulphur.

Q15. [SLO 2] Give two evidences that DNA is the genetic material.

Ans: DNA content is constant in somatic cells and half in gametes; mutations in DNA change characters; DNA is the transforming principle.

Q16. [SLO 3] Name the three models of DNA replication.

Ans: Semi-conservative, conservative and dispersive.

Q17. [SLO 3] What is semi-conservative replication?

Ans: Each daughter DNA contains one old parental strand and one newly synthesized strand.

Q18. [SLO 4] What did Meselson and Stahl use?

Ans: E. coli grown in heavy nitrogen (^{15}N) then light ^{14}N, with CsCl density gradient centrifugation.

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Q19. [SLO 4] How did the Meselson-Stahl result rule out the conservative and dispersive models?

Ans: First generation gave one intermediate band (not heavy + light = not conservative); second generation gave hybrid and light bands (not one intermediate band = not dispersive).

Q20. [SLO 5] What is the role of helicase, primase and DNA ligase?

Ans: Helicase unwinds DNA; primase makes RNA primer; ligase joins Okazaki fragments.

Q21. [SLO 5] What are Okazaki fragments?

Ans: Short DNA pieces made discontinuously on the lagging strand.

Q22. [SLO 5] Differentiate leading and lagging strand.

Ans: Leading: continuous synthesis towards the fork. Lagging: discontinuous Okazaki fragments away from the fork.

Q23. [SLO 5] Why can DNA polymerase add nucleotides only in 5' to 3' direction?

Ans: It needs a free 3'-OH of the preceding nucleotide to add the next one.

Q24. [SLO 5] What is the role of DNA polymerase I and III?

Ans: Polymerase III synthesizes new strands (with proofreading); polymerase I removes RNA primers and fills gaps with DNA.

Q25. [SLO 5] What is replication origin and replication fork?

Ans: Origin: site where replication begins. Fork: Y-shaped region where strands separate and are copied.

Q26. [SLO 6] How is DNA both stable and variable?

Ans: Accurate replication (base pairing, proofreading, repair) gives stability; rare errors and mutations give variability.

Q27. [SLO 7] Define genetic code and codon.

Ans: Genetic code: rules for translating nucleotide sequence into amino acids. Codon: three mRNA bases coding for one amino acid.

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Q28. [SLO 7] How many codons are there? How many code for amino acids?

Ans: 64; 61 for amino acids and 3 stop codons.

Q29. [SLO 7] Name the start and stop codons.

Ans: Start: AUG (methionine). Stop: UAA, UAG, UGA.

Q30. [SLO 7] What are the features of the genetic code?

Ans: Triplet, non-overlapping, comma-less, degenerate, unambiguous and universal.

Q31. [SLO 7] What is degeneracy of the code?

Ans: More than one codon can code for the same amino acid (e.g. six for leucine).

Q32. [SLO 8] Differentiate genetic code and codon; codon and anticodon.

Ans: Code: the whole system. Codon: triplet on mRNA; anticodon: complementary triplet on tRNA.

Q33. [SLO 8] Who deciphered the genetic code?

Ans: Nirenberg and Matthaei (poly-U = Phe), Khorana and Holley (Nobel 1968).

Q34. [SLO 9] Define transcription. Which enzyme performs it?

Ans: Synthesis of RNA from a DNA template by RNA polymerase.

Q35. [SLO 9] What is a promoter?

Ans: DNA region where RNA polymerase binds to start transcription (TATA box, CAAT box).

Q36. [SLO 9] Which strand is the template?

Ans: The antisense strand is read; the sense (coding) strand has the same sequence as mRNA except T for U.

Q37. [SLO 9] What are the three stages of transcription?

Ans: Initiation, elongation and termination.

Q38. [SLO 9] What is the role of sigma factor?

Ans: It helps the prokaryotic RNA polymerase recognize and bind the promoter.

Q39. [SLO 9] Name RNA polymerases of eukaryotes and their products.

Ans: Pol I: rRNA; Pol II: mRNA; Pol III: tRNA.

Q40. [SLO 10] What are exons and introns?

Ans: Exons: coding sequences retained in mature mRNA. Introns: non-coding sequences removed by splicing.

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Q41. [SLO 10] Why is eukaryotic mRNA shorter than its gene?

Ans: Introns are cut out and exons joined during processing (splicing).

Q42. [SLO 10] What are the 5' cap and poly-A tail for?

Ans: They protect mRNA from degradation and help its export and translation.

Q43. [SLO 11] Define translation.

Ans: The synthesis of a polypeptide on the ribosome according to the codons of mRNA.

Q44. [SLO 11] Name the three sites of the ribosome.

Ans: A (aminoacyl), P (peptidyl) and E (exit) sites.

Q45. [SLO 11] What is the role of tRNA and aminoacyl-tRNA synthetase?

Ans: tRNA brings amino acids and reads codons with its anticodon; the synthetase attaches the correct amino acid to its tRNA using ATP.

Q46. [SLO 11] How does a polypeptide chain terminate?

Ans: When a stop codon enters the A site, release factors free the chain.

Q47. [SLO 11] What is a polysome?

Ans: Several ribosomes translating the same mRNA at once.

Q48. [SLO 12] Differentiate prokaryotic and eukaryotic translation.

Ans: Prokaryotic: 70S, fMet, coupled with transcription, polycistronic mRNA. Eukaryotic: 80S, Met, after mRNA processing, monocistronic.

Q49. [SLO 13] What happens to a polypeptide after synthesis?

Ans: It folds (with chaperones), may be modified, and is sent to its destination (cytosol, ER, Golgi, secretion).

Q50. [SLO 14] Why is regulation of gene expression important?

Ans: Cells make only the proteins needed, at the right time and place; saves energy and allows differentiation.

Q51. [SLO 14] What is an operon?

Ans: A group of structural genes with a common promoter and operator, controlled together.

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Q52. [SLO 15] What is negative control in the lac operon?

Ans: Without lactose a repressor binds the operator and blocks transcription; lactose inactivates the repressor and switches the operon on.

Q53. [SLO 15] What are the components of the lac operon?

Ans: Regulator i gene, promoter, operator, and structural genes z, y, a.

Q54. [SLO 15] What is an inducer? Name the one in lac operon.

Ans: A molecule that switches the operon on; lactose (allolactose) binds the repressor.

Q55. [SLO 16] What is positive control of gene expression?

Ans: An activator (CAP-cAMP) binds DNA and increases transcription when glucose is low.

Q56. [SLO 17] Define mutation and mutagen.

Ans: Mutation: sudden heritable change in DNA or chromosomes. Mutagen: agent that causes mutation.

Q57. [SLO 17] Name the sources of mutations.

Ans: Replication errors and background radiation (natural); UV, X-rays, chemicals like nitrous acid and mustard gas (induced).

Q58. [SLO 18] Differentiate natural and induced mutations.

Ans: Natural: spontaneous, rare. Induced: caused by mutagens, frequent.

Q59. [SLO 19] Why are most mutations harmful?

Ans: A random change in an already well-adapted gene mostly decreases protein function.

Q60. [SLO 20] How do mutations contribute to evolution?

Ans: They create new alleles; beneficial ones are selected by natural selection and spread.

Q61. [SLO 21] What is non-disjunction?

Ans: Failure of homologous chromosomes or chromatids to separate in meiosis, giving gametes with extra or missing chromosomes.

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Q62. [SLO 21] Differentiate trisomy and monosomy.

Ans: Trisomy: an extra chromosome (2n + 1, Down syndrome). Monosomy: one chromosome missing (2n - 1, Turner syndrome).

Q63. [SLO 21] Write the cause and features of Down, Klinefelter and Turner syndromes.

Ans: Down: trisomy 21 (47). Klinefelter: XXY male, sterile. Turner: XO female (45), short and sterile.

Q64. [SLO 22] What is a point mutation? Name its types.

Ans: Change in a single nucleotide: substitution, insertion or deletion.

Q65. [SLO 22] What is a frameshift mutation?

Ans: Insertion or deletion of bases (not multiple of three) that shifts the reading frame and changes all later amino acids.

Q66. [SLO 22] What causes sickle cell anaemia and PKU?

Ans: Sickle cell: substitution GAG to GUG (glutamic acid to valine) in beta-globin. PKU: defective phenylalanine hydroxylase gene.

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Chapter 13: Long Questions with Answers (SLO based)

Q1. [SLO 1] Describe the structure of a chromosome and the packaging of DNA.

Ans:

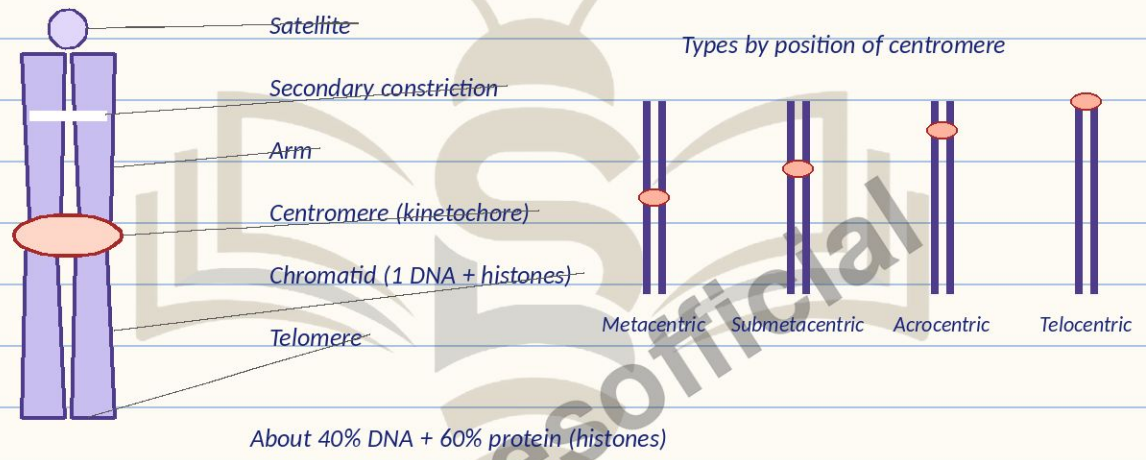


Fig: Structure and types of chromosome

- *Chromosome = DNA + histone protein (about 40% DNA); two chromatids joined at the centromere (kinetochore); arms; telomeres; secondary constriction and satellite. Types: metacentric, submetacentric, acrocentric, telocentric.*
- *Packaging: DNA wraps twice round 8 histones (nucleosome, 10 nm) to chromatin fibre (30 nm), supercoil (300 nm), chromatid (700 nm), chromosome (1400 nm). Euchromatin active, heterochromatin inactive.*

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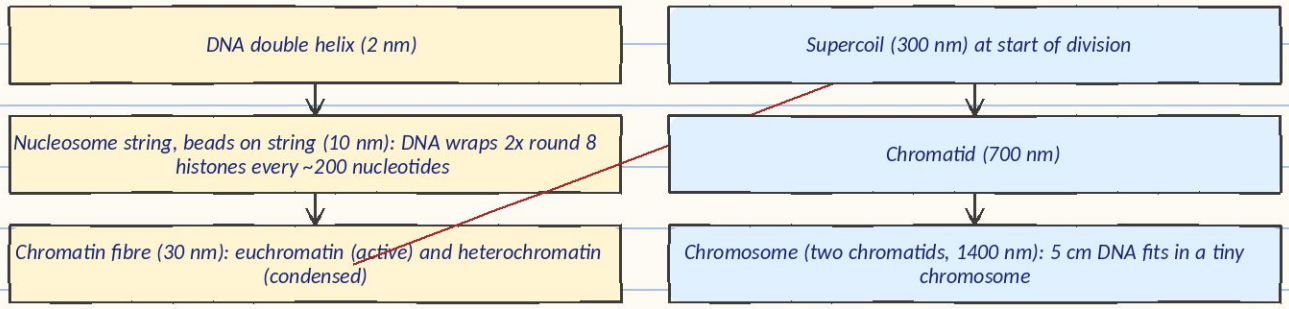
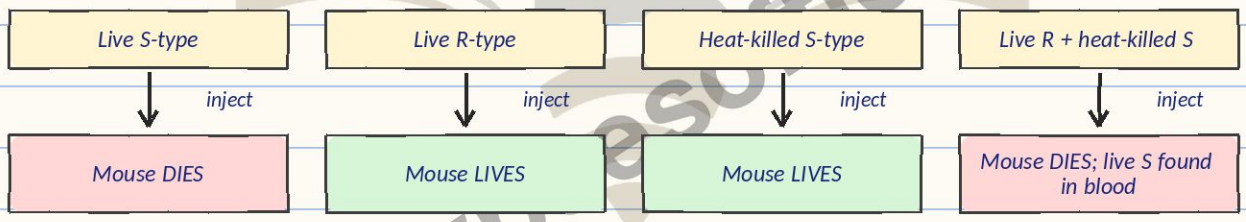


Fig: Levels of organization of a chromosome

Q2. [SLO 2] Describe the experiments that proved DNA is the genetic material.

Ans:



S-type: smooth, polysaccharide capsule, virulent; R-type: rough, no capsule, non-virulent

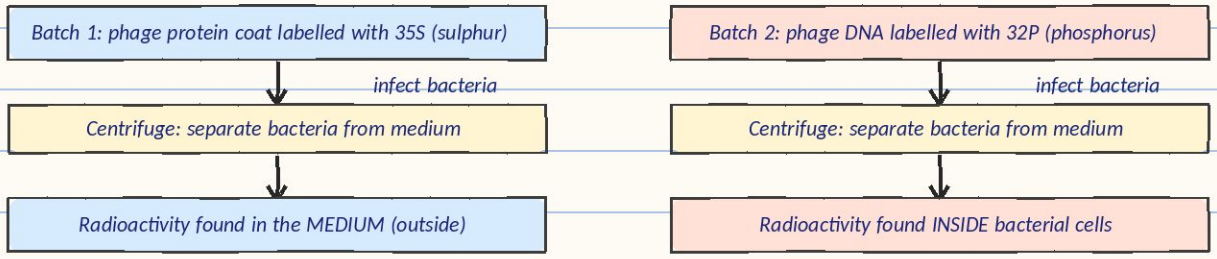
Transformation: heat-killed S passed genetic material to live R (turned it into S)

Avery, Macleod, McCarty (1944): only DNAase stops transformation = DNA is the transforming principle

Fig: Griffith's experiment (1928) on Streptococcus pneumoniae

- *Griffith (1928): live S kills; live R and heat-killed S are harmless; live R + heat-killed S kills and live S is recovered: transformation. Avery, MacLeod and McCarty (1944): DNA is the transforming principle.*

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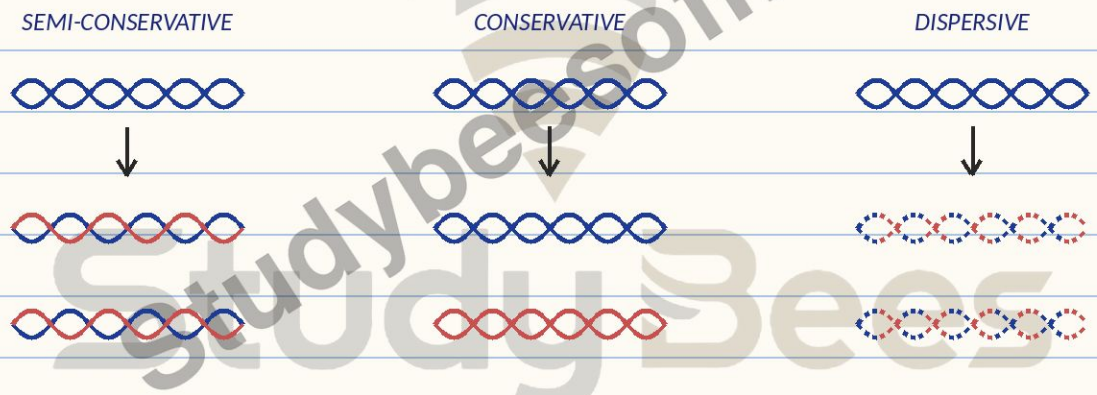
Conclusion: DNA (not protein) enters the cell and directs production of new viruses (1952)

Fig: Hershey and Chase experiment on bacteriophage

- *Hershey and Chase (1952): ^{35}S protein stayed outside, ^{32}P DNA entered bacteria and produced new phages: DNA is the genetic material.*

Q3. [SLO 3, 4] Describe the three models of DNA replication and the Meselson-Stahl experiment.

Ans:



Blue = parental strand; red = newly made strand

Meselson and Stahl (1958) proved the semi-conservative model

Fig: Three models of DNA replication

- *Models: semi-conservative (one old + one new strand), conservative (old DNA stays whole), dispersive (mixed fragments).*

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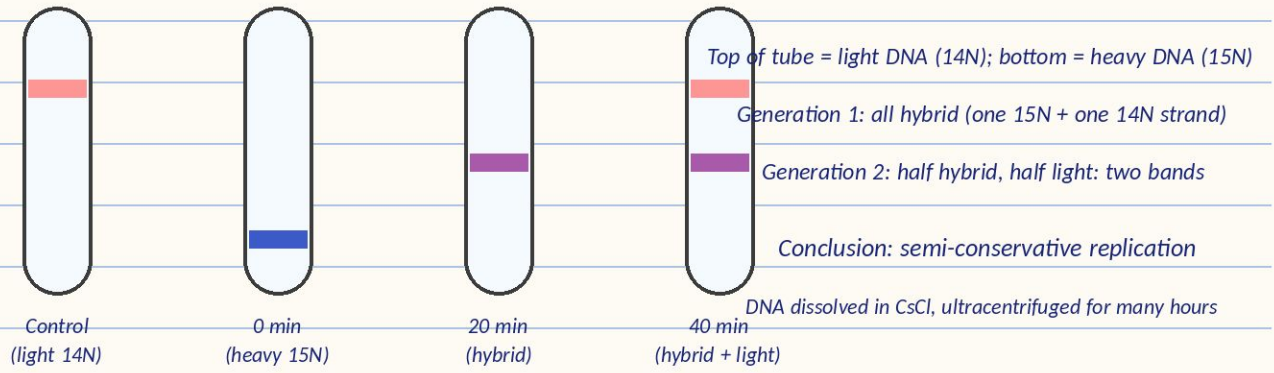
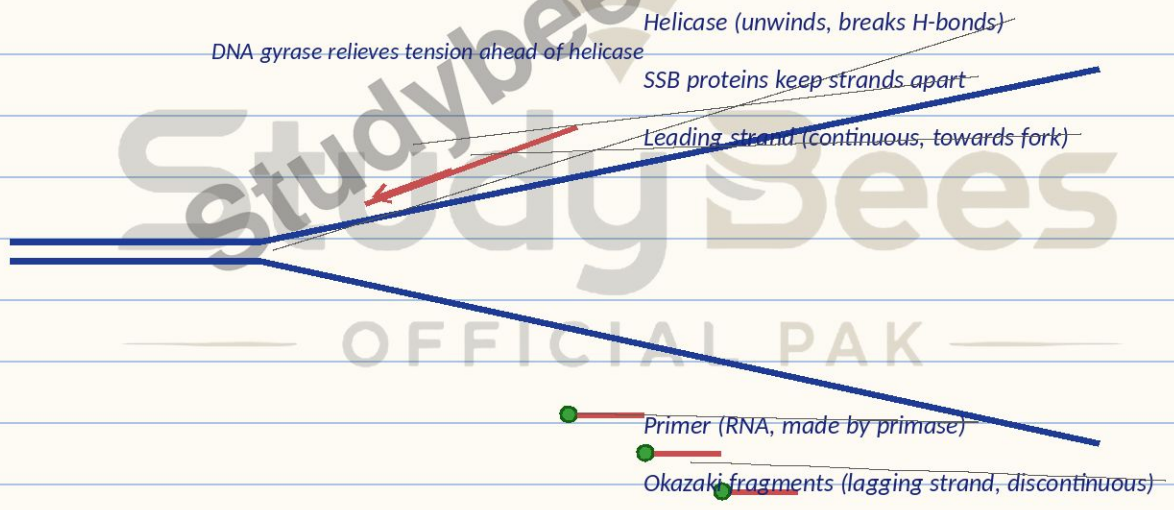


Fig: Meselson-Stahl experiment (heavy nitrogen 15N shifted to 14N medium)

- *Meselson-Stahl (1958): E. coli from ^{15}N to ^{14}N, CsCl gradient: generation 1 one hybrid band (rules out conservative); generation 2 hybrid + light bands (rules out dispersive): semi-conservative.*

Q4. [SLO 5, 6] Describe the process of DNA replication. Explain how DNA is stable yet variable.

Ans:



DNA polymerase III: adds nucleotides 5' to 3' on 3'-OH of primer; proofreading

DNA polymerase I replaces primers by DNA; DNA ligase joins fragments (phosphodiester bonds)

Fig: Replication fork

- *Origin, helicase unwinding, SSB, primase, DNA polymerase III elongation (leading continuous, lagging as Okazaki fragments), polymerase I primer removal, ligase, proofreading. Stable by pairing and proofreading; variable by*

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rare errors and mutation.

Q5. [SLO 7, 8] Describe the genetic code and its characteristics.

Ans:

- *64 codons: 61 sense and 3 stop (UAA, UAG, UGA); start AUG. Triplet, non-overlapping, comma-less, degenerate, unambiguous, universal. Deciphered by Nirenberg, Khorana, Holley.*

Q6. [SLO 9, 10] Describe transcription and processing of mRNA in eukaryotes.

Ans:

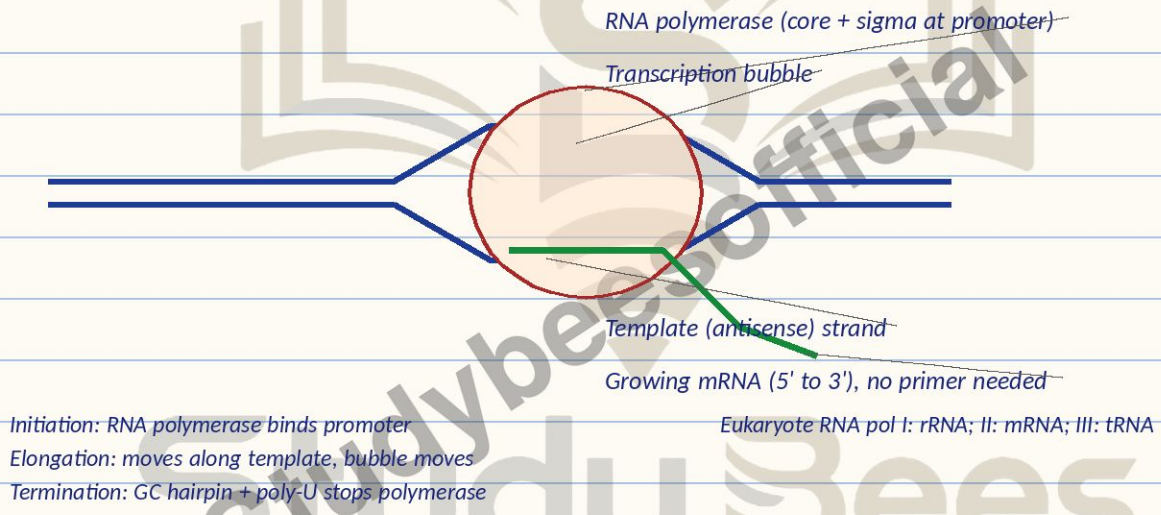


Fig: Transcription (synthesis of RNA from DNA)

- *Transcription: RNA polymerase binds the promoter (initiation), adds ribonucleotides 5' to 3' on the template strand (elongation), stops at terminator (termination); no primer.*

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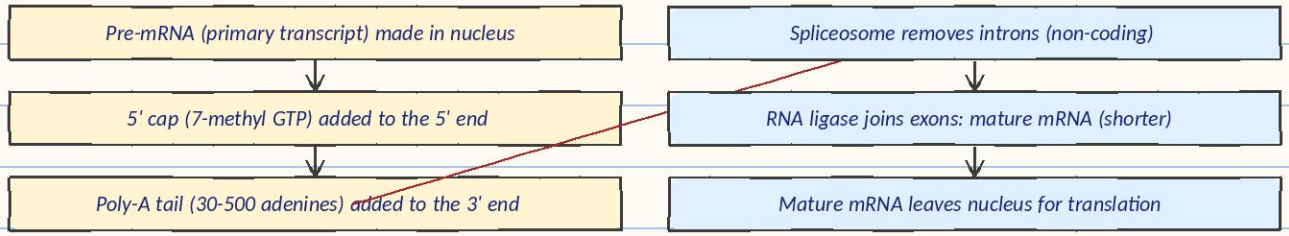
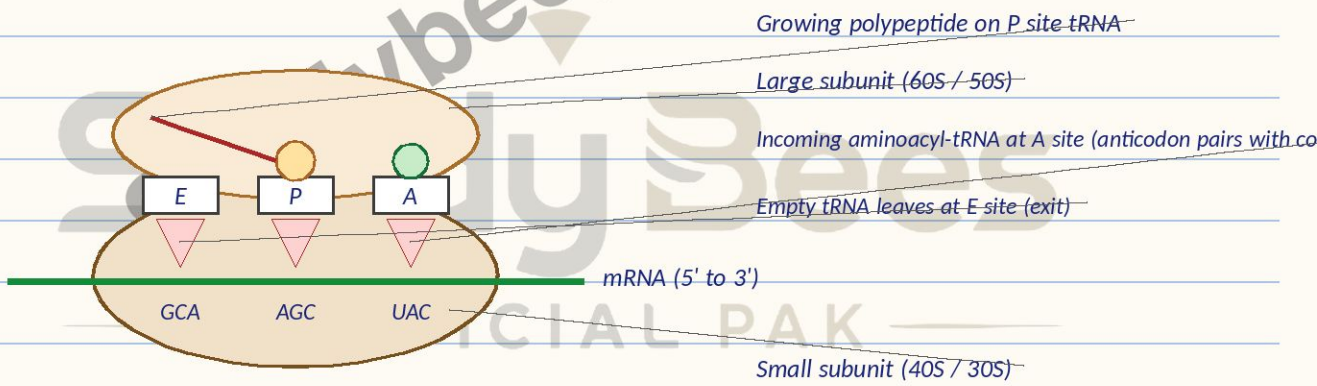


Fig: mRNA processing in eukaryotes (why mRNA shortens)

- *Eukaryotic pre-mRNA gets a 5' cap and poly-A tail; spliceosome removes introns and joins exons, so mature mRNA is shorter and leaves for the cytoplasm.*

Q7. [SLO 11, 12] Describe protein synthesis (translation) and compare prokaryotes and eukaryotes.

Ans:



Elongation steps: (a) aminoacyl-tRNA enters A site; (b) peptidyl transferase makes peptide bond; (c) translocation moves ribosome 5' to 3' (A to P, P to E). Stop codon UAG, UAA or UGA: release factors.

Fig: Translation - polypeptide elongation on the ribosome

- *Initiation (small subunit + AUG + Met-tRNA, large subunit joins); elongation (aminoacyl-tRNA at A site, peptide bond by peptidyl transferase, translocation); termination at UAA, UAG or UGA with release factors.*

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Prokaryote	Eukaryote
70S, fMet, coupled with transcription	80S, Met, after processing
Polycistronic mRNA	Monocistronic mRNA

Q8. [SLO 13] Describe the fate of a newly made polypeptide.

Ans:

- Folding by chaperones; modification; free ribosome proteins stay in cytosol / nucleus / mitochondria; rough ER ribosome proteins enter ER, then Golgi, vesicles to secretion, membrane or lysosomes.

Q9. [SLO 14-16] Explain regulation of gene expression with the lac operon: negative and positive control.

Ans:

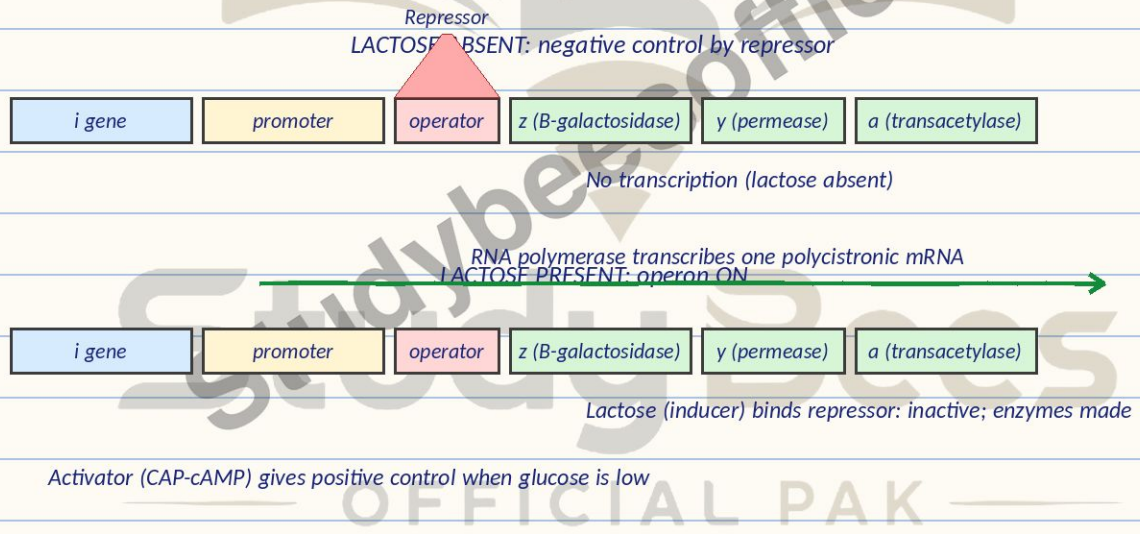


Fig: Lac operon (positive and negative control of gene expression)

- Negative: repressor on operator blocks transcription; lactose (inducer) inactivates it. Positive: CAP-cAMP enhances transcription when glucose is low.

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Q10. [SLO 17, 18] Define mutation; describe sources of mutation and differentiate natural and induced mutations.

Ans:

- Mutation: sudden heritable change. Natural: replication errors, background radiation. Induced: UV, X-rays, gamma rays, nitrous acid, mustard gas.

Natural	Induced
Spontaneous, rare	By mutagens, frequent

Q11. [SLO 19, 20] Why are most mutations harmful? Explain the role of mutations in evolution.

Ans:

- Most are harmful because random change disturbs a well-adapted protein; some are neutral; a few beneficial.
- Mutations produce new alleles; natural selection favours beneficial ones (antibiotic resistance); the basis of evolution and of plant breeding.

Q12. [SLO 21] Describe chromosomal mutations and the syndromes caused by non-disjunction.

Ans:

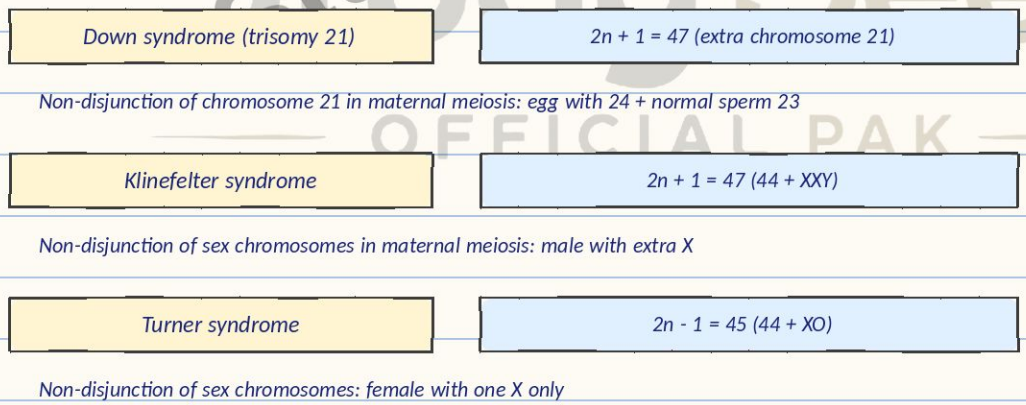


Fig: Chromosomal mutations through non-disjunction

- Structural: deletion, duplication, inversion, translocation. Numerical: aneuploidy (non-disjunction), polyploidy.

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Q13. [SLO 22] Describe gene (point) mutations with examples.

Ans:

Normal mRNA	AUG AGC GCA UUC	Met - Ser - Ala - Phe
Substitution	AUG AGC GCU UUC	Met - Ser - Ala - Phe (silent) or changed amino acid
Insertion	AUG AGC AGC AUU C	Frameshift: all codons after change
Deletion	AUG AGC CAU UC	Frameshift: non-functional protein

Fig: Types of point mutation (sample sequence)

Point mutation = base substitution, insertion or deletion; frameshift mutation = insertion / deletion not in multiples of three

- Substitution (silent, missense, nonsense), insertion, deletion (frameshift).

  Examples: sickle cell (GAG to GUG), PKU, albinism, haemophilia, cystic fibrosis.

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